EE6497 – Homework 1

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Question 1

(a)

Taking the exponential of both sides gives

$$p(x) = e^{ax+b} = e^b e^{ax}, \qquad x \geq 0.$$

Since $p(x)$ is a probability density,

$$1 = \int_0^\infty e^{ax+b}\, dx = -\frac{e^b}{a}.$$

Thus $e^b = -a$. With $\lambda = -a > 0$, this becomes

$$\boxed{p(x) = \lambda e^{-\lambda x} = \mathrm{Exp}(x \mid \lambda), \quad \lambda = -a.}$$

It is an exponential distribution with rate $-a$, and $b = \log(-a)$.

(b)

Yes. The variable z selects one of three exponential distributions. Summing over its possible values,

$$\begin{aligned} p(x) &= \sum_{k=1}^{3} p(z=k)p(x \mid z=k) \\ &= \boxed{0.5\,\mathrm{Exp}(x \mid 7.8) + 0.4\,\mathrm{Exp}(x \mid 0.6) + 0.1\,\mathrm{Exp}(x \mid 4.5).} \end{aligned}$$

The weights are nonnegative and sum to one, so this is a valid mixture model.

(c)

Let $\mathrm{SSE} = \sum_i (y_i - \hat{y}_i)^2$ and $\mathrm{SST} = \sum_i (y_i - \bar{y})^2$. Assuming $\mathrm{SST} > 0$,

$$R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}} < 0 \quad \Longleftrightarrow \quad \mathrm{SSE} > \mathrm{SST}.$$

Thus R^2 can be negative when the predictions have a larger squared error than simply predicting the mean $\bar{y}$ for every observation.

For example, take $y = (1, 3)$ and $\hat{y} = (0, 0)$. Then $\bar{y} = 2$, $\mathrm{SST} = 2$ and $\mathrm{SSE} = 10$, giving $R^2 = 1 - 10/2 = -4$.

For ordinary least squares with an intercept, fitted optimally and evaluated on the same training data, R^2 cannot be negative: the constant mean predictor is already an available choice. This restriction does not hold for arbitrary predictions or evaluation on a different dataset. If $\mathrm{SST} = 0$, the stated formula is undefined.

Question 2

(a)(i)

For a given state pair, the joint probability factors as

$$p(z[1], z[2], o_2, o_1) = \pi(z[1])p(o_2 \mid z[1])T(z[1], z[2])p(o_1 \mid z[2]).$$

Using the initial probability $1/2$ for either state gives:

$z[1]$	$z[2]$	Joint probability
s_1	s_1	$\frac{1}{2}(1-p)ap$
s_1	s_2	$\frac{1}{2}(1-p)(1-a)q$
s_2	s_1	$\frac{1}{2}(1-q)(1-b)p$
s_2	s_2	$\frac{1}{2}(1-q)bq$

(a)(ii)

When the states are hidden, all four state paths can produce the observed sequence. Therefore their joint probabilities must be added:

$$\begin{aligned} L(a,b) &= p(o_2, o_1) \\ &= \frac{1}{2}\big[(1-p)ap + (1-p)(1-a)q \\ &\qquad + (1-q)(1-b)p + (1-q)bq\big] \\ &= \boxed{\frac{1}{2}\big[(1-p)\{ap + (1-a)q\} + (1-q)\{(1-b)p + bq\}\big].} \end{aligned}$$

(b)(i)

The observed transition counts are

$$N_{11} = 1, \quad N_{12} = 1, \quad N_{21} = 0, \quad N_{22} = 2.$$

Assuming $0 < p, q < 1$, independence of the four sequences gives

$$L(a,b) = C\, a(1-a)b^2, \qquad C = \frac{1}{16}(1-p)^2 p\, q^3 (1-q)^2 > 0.$$

The constant C does not depend on a or b. Since

$$a(1-a) = \frac{1}{4} - \left(a - \frac{1}{2}\right)^2,$$

it is maximized at $a = 1/2$. Also, b^2 is maximized at $b = 1$ on $[0, 1]$. Hence

$$\boxed{\hat{a}_{\mathrm{MLE}} = \frac{1}{2}, \qquad \hat{b}_{\mathrm{MLE}} = 1.}$$

These are also the observed proportions of self-transitions from each state. If p or q equals 0 or 1, some of the given records are impossible under the model, and the full likelihood is zero for every (a, b); the above unique estimate then does not apply.

(b)(ii)

Here only one emission sequence is observed. Expanding the result in (a)(ii),

$$L(a,b) = C_0 + Aa + Bb,$$

where

$$\begin{aligned} C_0 &= \tfrac{1}{2}\big[(1-p)q + (1-q)p\big], \\ A &= \tfrac{1}{2}(1-p)(p-q), \qquad B = \tfrac{1}{2}(1-q)(q-p). \end{aligned}$$

For $0 < p, q < 1$, the signs of these coefficients determine the maximum over $0 \le a, b \le 1$:

- If $p > q$, then $A > 0$ and $B < 0$. Increasing a and decreasing b increases the likelihood, so $(\hat{a}, \hat{b}) = (1, 0)$.
- If $p < q$, then $A < 0$ and $B > 0$. The likelihood is maximized at $(\hat{a}, \hat{b}) = (0, 1)$.

Thus

$$\boxed{(\hat{a}_{\mathrm{MLE}}, \hat{b}_{\mathrm{MLE}}) = \begin{cases} (1,0), & p > q, \\ (0,1), & p < q. \end{cases}}$$

Intuitively, the optimal transition sends the second state to whichever state is more likely to emit o_1.

At the emission boundaries, these displayed choices remain maximizers but may not be unique. If $p = 1 > q$, then $\hat{b} = 0$ and any $a \in [0, 1]$ is optimal. If $q = 1 > p$, then $\hat{a} = 0$ and any $b \in [0, 1]$ is optimal.

(c)

I would use the **Baum–Welch algorithm**, which applies EM to an HMM. The states are unobserved, so their transition counts cannot be read directly from the data. Each length-20 sequence has 2^{20} possible state paths, whereas forward–backward computes the required posterior probabilities efficiently.

Starting from an initial choice of a, b, each iteration has two steps:

1. **E-step:** Run forward–backward on each of the five sequences using the current parameters. Compute the posterior expected transition counts and add them across sequences.
2. **M-step:** Update the self-transition probabilities by

$$a^{\mathrm{new}} = \frac{\mathbb{E}[N_{11}]}{\mathbb{E}[N_{11}] + \mathbb{E}[N_{12}]}, \qquad b^{\mathrm{new}} = \frac{\mathbb{E}[N_{22}]}{\mathbb{E}[N_{21}] + \mathbb{E}[N_{22}]}.$$

The expectations are conditional on the observed sequences and current parameters. There are $5(20-1) = 95$ transitions in total; no transition is counted between different sequences. The known p, q and initial probabilities remain fixed.

Repeat until the log likelihood changes negligibly. Exact EM updates do not decrease the observed-data likelihood, but a global maximum is not guaranteed. Different initial values can be tried, retaining the result with the highest likelihood.

Question 3

(a)

Since the devices are independent and $\sum_{k=1}^{r+1} n_k = n$, the likelihood is

$$L(\theta) = \prod_{k=1}^{r} \left[\theta^{k-1}(1-\theta)\right]^{n_k} (\theta^r)^{n_{r+1}}.$$

The final category contributes no factor of $1-\theta$. Define

$$S = \sum_{k=1}^{r}(k-1)n_k + rn_{r+1} = \sum_{k=1}^{r+1} kn_k - n, \qquad F = n - n_{r+1}.$$

Then $L(\theta) = \theta^S(1-\theta)^F$, and for $0 < \theta < 1$,

$$\boxed{\ell(\theta) = \left(\sum_{k=1}^{r+1} kn_k - n\right)\log\theta + (n - n_{r+1})\log(1-\theta).}$$

Boundary values are interpreted directly from the likelihood or by limits.

(b)

First suppose $S > 0$ and $F > 0$. Differentiating and setting the derivative to zero gives

$$\ell'(\theta) = \frac{S}{\theta} - \frac{F}{1-\theta} = 0 \quad \Longrightarrow \quad S(1-\theta) = F\theta \quad \Longrightarrow \quad \hat{\theta} = \frac{S}{S+F}.$$

Also,

$$\ell''(\theta) = -\frac{S}{\theta^2} - \frac{F}{(1-\theta)^2} < 0,$$

so this is the unique maximum. Since

$$S + F = \sum_{k=1}^{r+1} kn_k - n + n - n_{r+1} = \sum_{k=1}^{r+1} kn_k - n_{r+1},$$

we obtain

$$\boxed{\hat{\theta}_{\text{MLE}} = \frac{\displaystyle\sum_{k=1}^{r+1} kn_k - n}{\displaystyle\sum_{k=1}^{r+1} kn_k - n_{r+1}}.}$$

For completeness, the boundary cases are as follows (assuming $n \geq 1$):

- If all observations equal $r+1$, then $F = 0$, $S = rn$, and $L(\theta) = \theta^{rn}$. Its maximum is at the allowed endpoint $\hat{\theta} = 1$, consistent with the formula.
- If all observations equal 1, then $S = 0$ and $L(\theta) = (1-\theta)^n$. The formula gives 0, but the parameter space excludes 0. The likelihood approaches its supremum of 1 as $\theta \to 0^+$, without attaining it. Thus an MLE does not exist for this sample under $0 < \theta \leq 1$.